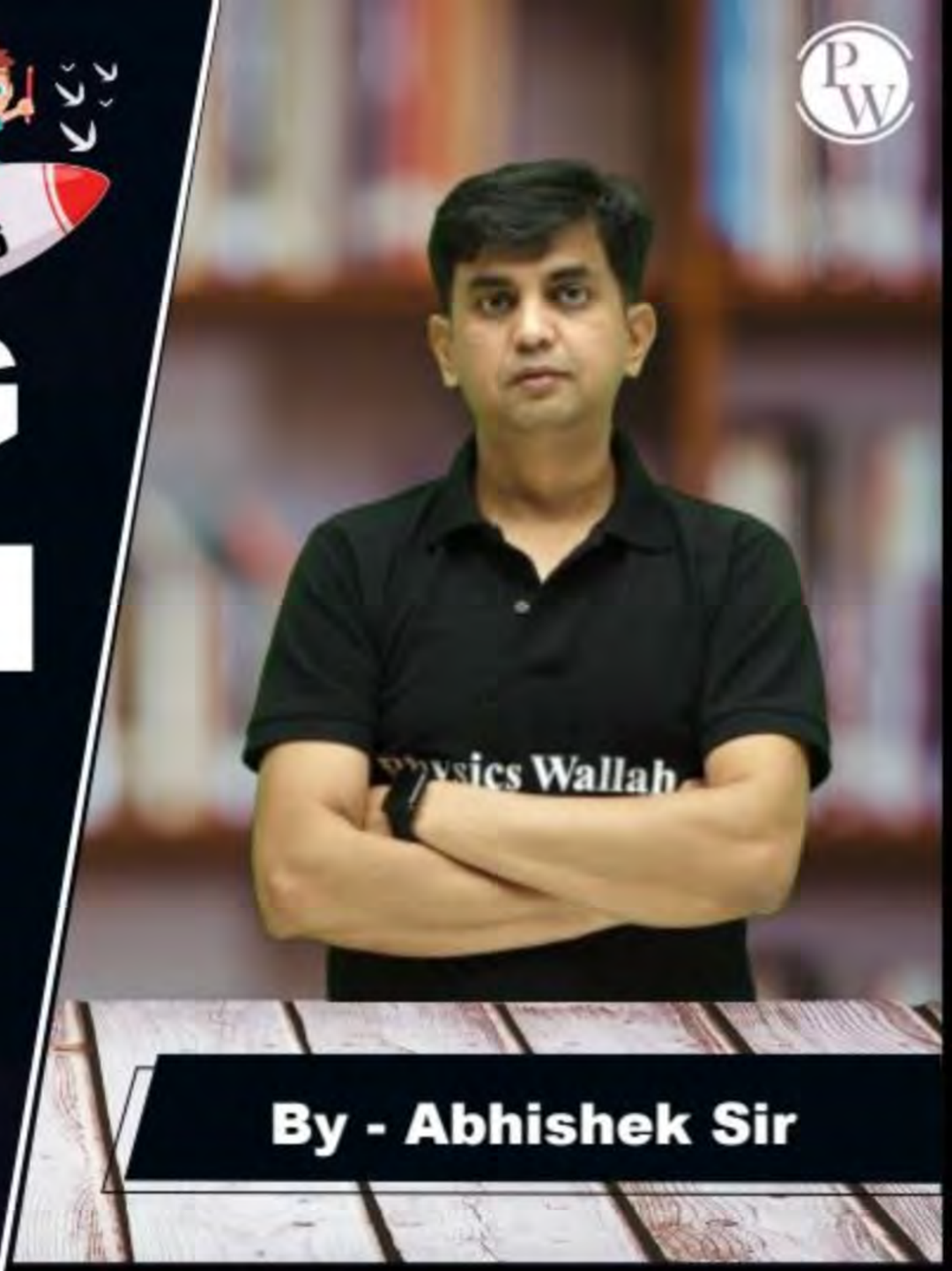
CS & IT ENGINEERING

Computer Network - 2

Framing & Error Control

Lecture No. - 03



By - Abhishek Sir

Recap of Previous Lecture

Topic: Bit Stuffing

Topic: Two-dimension (2D) Parity

Topics to be Covered

Topic Two-dimension (2D) Parity

Topic Hamming Code

[MCQ] [GATE-2014][1 Mark]

#Q. A bit-stuffing based framing protocol uses an 8-bit delimiter pattern of 01111110. If the output bit-string after stuffing is 01111100101, then the input bit-string is :

A. 0111110100

B. 0111110101

C. 0111111101

D. 0111111111

Ans: B

Frame delimiter = 01111110

Output = 01111100101

Transmitted data = 01111110 01111100101 01111110

SFD(Flag) discard EFD(Flag)

Input (Frame) = 011111101010 ~~01111110~~

(Recv At)

Ans

[MCQ] [GATE-IT-2004] [1 Mark]

#Q. In a data link protocol, the frame delimiter flag is given by 0111. Assuming that bit stuffing is employed, the transmitter sends the data sequence 01110110 as :

A 01101011

B 011010110

C 011101100

D 0110101100

Ans: D

Frame delimiter = 0111 / 01110

Input (Frame) = 01110110

Transmitted data = 01110 0110101100 01110

SFD

EFD

Output = 0110101100

Output = Input with bit stuffing

Topic : Two-dimension (2D) Parity

→ Both sender and receiver must agree on same block size and same parity (either even or odd parity)

→ Block Size = m * n
[m rows & n columns]

[No. of parity bits is depend on no of data bits]

Number of Data bits = (m–1) * (n–1)

Number of parity bits = [m + n – 1]

⇒ Single bit Error correction

Topic : Two-dimension (2D) Parity

$R_1 = d_1 \oplus d_2 \oplus d_3 \oplus d_4 \oplus d_5$
$R_2 = d_6 \oplus d_7 \oplus d_8 \oplus d_9 \oplus d_{10}$
$R_3 = d_{11} \oplus d_{12} \oplus d_{13} \oplus d_{14} \oplus d_{15}$

$C_1 = d_1 \oplus d_6 \oplus d_{11}$
$C_2 = d_2 \oplus d_7 \oplus d_{12}$
$C_3 = d_3 \oplus d_8 \oplus d_{13}$
$C_4 = d_4 \oplus d_9 \oplus d_{14}$
$C_5 = d_5 \oplus d_{10} \oplus d_{15}$

$P = R_1 \oplus R_2 \oplus R_3 = C_1 \oplus C_2 \oplus C_3 \oplus C_4 \oplus C_5$

Suppose Block Size = 4 x 6 and "Even Parity"

Data = "$d_1 d_2 d_3 d_4 d_5\ d_6 d_7 d_8 d_9 d_{10}\ d_{11} d_{12} d_{13} d_{14} d_{15}$"

Sender

d_1	d_2	d_3	d_4	d_5	R_1
d_6	d_7	d_8	d_9	d_{10}	R_2
d_{11}	d_{12}	d_{13}	d_{14}	d_{15}	R_3
C_1	C_2	C_3	C_4	C_5	P

Transmitted Data = "$d_1 d_2 d_3 d_4 d_5 R_1\ d_6 d_7 d_8 d_9 d_{10} R_2\ d_{11} d_{12} d_{13} d_{14} d_{15} R_3\ C_1 C_2 C_3 C_4 C_5 P$"

Topic : Two-dimension (2D) Parity

CASE I : No any error

Suppose Block Size = 4 x 6 and "Even Parity"

Receiver

1	0	0	1	1	1
0	1	1	0	1	1
1	0	0	1	0	0
0	1	1	0	0	0

Transmitted Data = "1 0 0 1 1 1 0 1 1 0 1 1 1 0 0 1 0 0 0 1 1 0 0 0"

Received Data = "1 0 0 1 1 1 0 1 1 0 1 1 1 0 0 1 0 0 0 1 1 0 0 0"

Receiver Concluded : No any Error detected; Accept the data;

Topic : Two-dimension (2D) Parity

Error Dection :-

if receiver finds all row wise and column wise parities are balanced

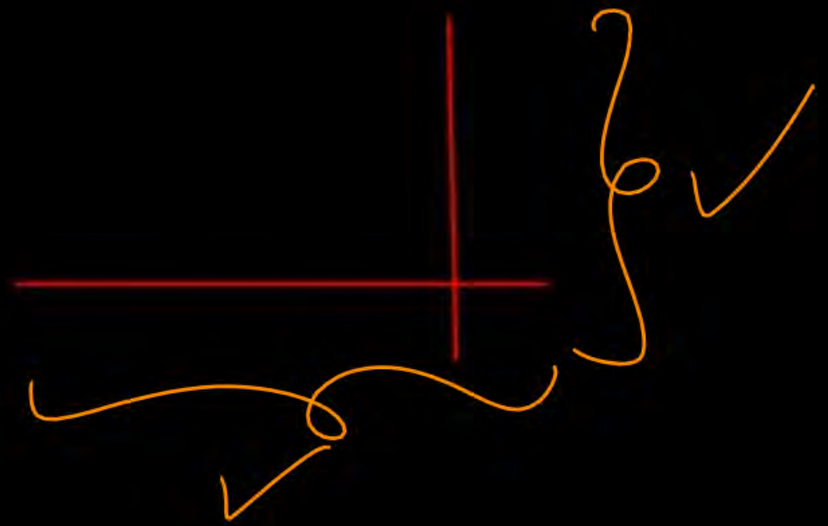
then receiver concluded "No any error detected"

else

receiver concluded "Error detected", call Error Correction;

Topic : Two-dimension (2D) Parity

CASE I : No any error



Topic : Two-dimension (2D) Parity

CASE II : One-bit error

Suppose Block Size = 4 x 6 and "Even Parity"

Receiver

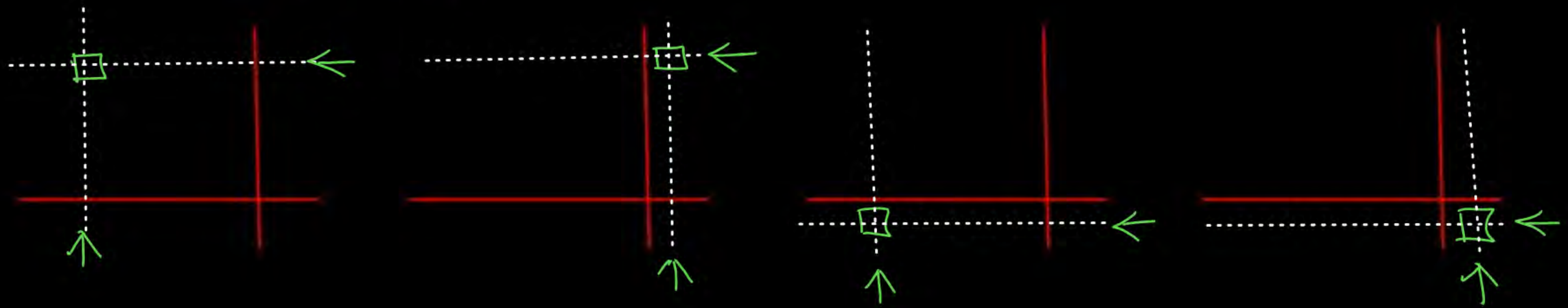
Transmitted Data = "1 0 0 1 1 1 0 1 1 0 1 1 1 0 0 1 0 0 0 1 1 0 0 0"

Received Data = "1 0 0 1 1 1 0 1 0 0 1 1 1 0 0 1 0 0 0 1 1 0 0 0"

Receiver Concluded : Error Detected; Error Correction: Single bit Error case ⇒ Error Corrected Successfully

Topic : Two-dimension (2D) Parity

CASE II : One-bit error $\Rightarrow$ Always detected and corrected successfully



Topic : Two-dimension (2D) Parity

CASE III : Two-bit error

Suppose Block Size = 4 x 6 and "Even Parity"

Receiver

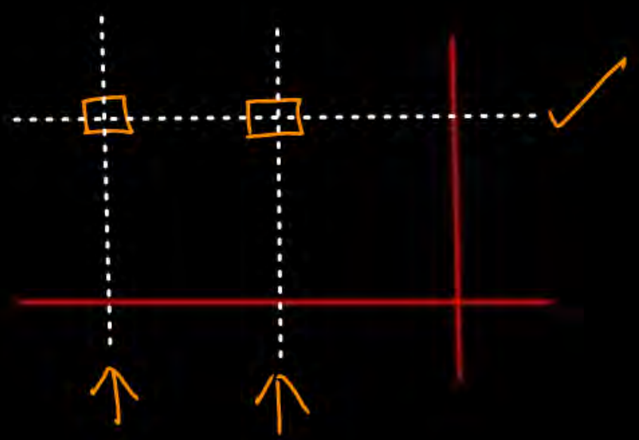
Transmitted Data = "1 0 0 1 1 1 0 1 1 0 1 1 1 0 0 1 0 0 0 1 1 0 0 0"

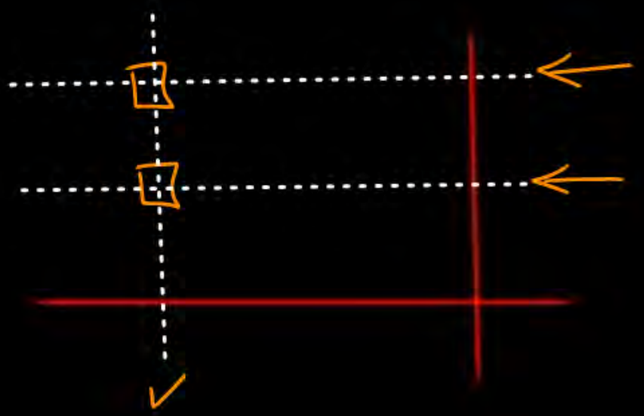
Received Data = "1 0 0 1 1 1 0 1 0 0 1 1 1 1 0 1 0 0 0 1 1 0 0 0"

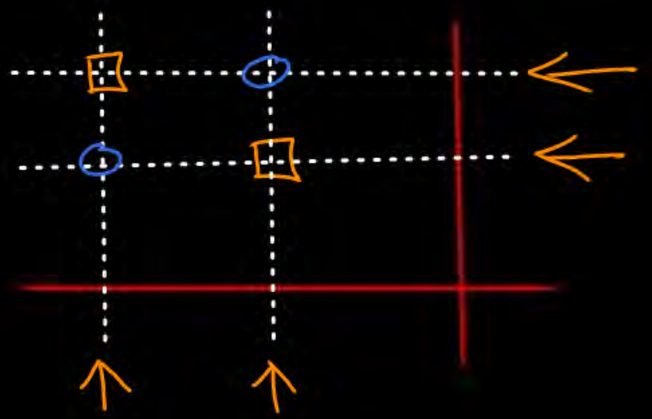
Receiver Concluded : Error Detected; Error Correction: Min^n two bit Error Unable to correct;

Topic : Two-dimension (2D) Parity

CASE III : Two-bit error → Always detected, Unable to correct

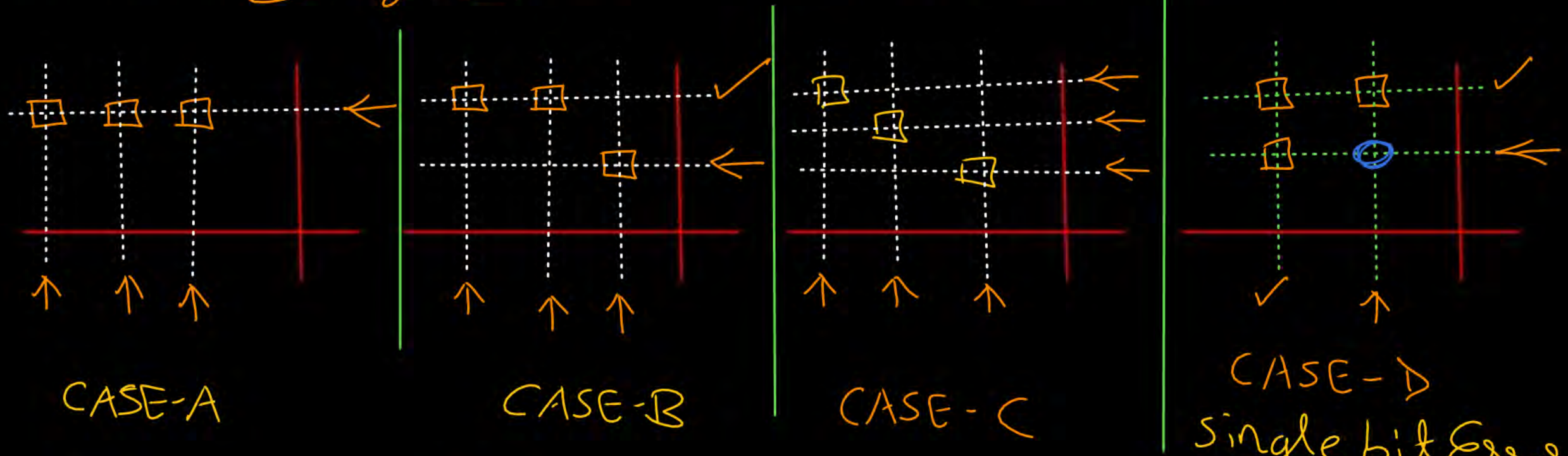






Topic : Two-dimension (2D) Parity

CASE IV : Three-bit error → Always detected, Unable to correct



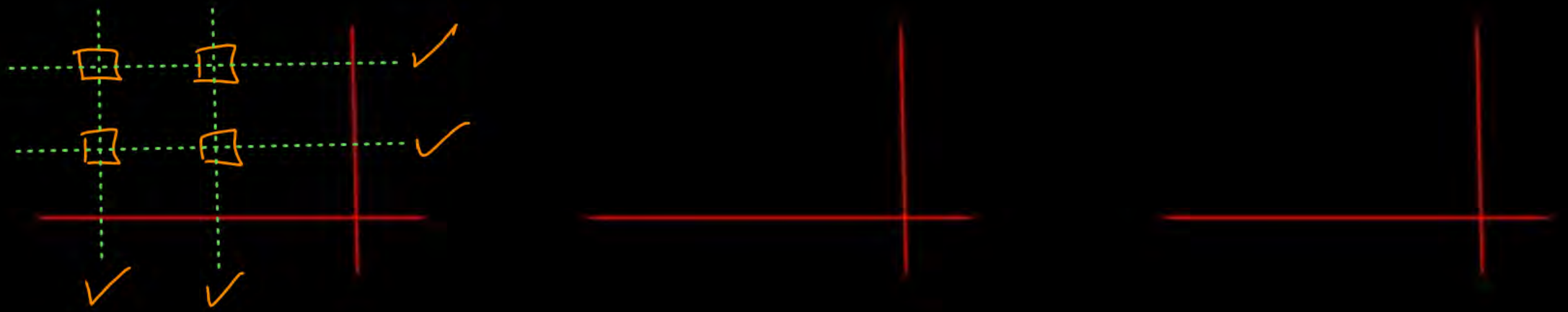
CASE-A CASE-B CASE-C CASE-D

CASE A, B & C ⇒ Min^m three bit error (Unable to correct)

CASE-D single bit Error Corrected successfully

Topic : Two-dimension (2D) Parity

CASE V : Four-bit error → May be detected



Example :-

#Q. Consider 2D parity scheme for error detection using the Even parity, the block size is 4 x 6 (including parity bits) contains 15 data bits. The matrix shows data received by a receiver and has n corrupted bits. What is the minimum possible value of n ?

(Even Parity)

1	1	0	1	1	1
0	1	1	0	1	1
1	0	0	0	0	0
0	1	1	0	0	0

Topic : Two-dimension (2D) Parity

Minimum number of bits corrupted in the Block

= Maximum [total number of row wise parity unbalanced,
total number of column wise parity unbalanced]

= Max[2, 2]
= 2

Ans = 2

[MCQ] IISC, H.W. [GATE-IT-2008] [2 Mark]

#Q. Data transmitted on a link uses the following 2D parity scheme for error detection :

Each sequence of 28 bits is arranged in a 4 x 7 matrix (rows r0 through r3, and columns d7 through d1) and is padded with a column d0 and row r4 of parity bits computed using the Even parity scheme. Each bit of column d0 (respectively, row r4) gives the parity of the corresponding row (respectively, column). These 40 bits are transmitted over the data link.

The table shows data received by a receiver and has n corrupted bits. What is the minimum possible value of n ?

	d_7	d_6	d_5	d_4	d_3	d_2	d_1	d_0
r_0	0	1	0	1	0	0	1	1
r_1	1	1	0	0	1	1	1	0
r_2	0	0	0	1	0	1	0	0
r_3	0	1	1	0	1	0	1	0
r_4	1	1	0	0	0	1	1	0

A 1

B 2

C 3

D 4

Topic : Two-dimension (2D) Parity

→ Receiver **detect** and **correct** "**all single bit error**"

→ In case of burst error,
receiver may able to detect "**burst error**"

$d_1\ d_2$	R_1
$d_3\ d_4$	R_2
$C_1\ C_2$	P

Topic : Two-dimension (2D) Parity

* Linear code

Two-dimension Parity (with Even Parity) and 4 data bits :

Data	→	Codeword	Data	→	Codeword
$d_1d_2d_3d_4$	→	$d_1d_2R_1d_3d_4R_2C_1C_2P$	$d_1d_2d_3d_4$	→	$d_1d_2R_1d_3d_4R_2C_1C_2P$
0 0 0 0	→	0 0 0 0 0 0 0 0 0 0	1 0 0 0	→	1 0 1 0 0 0 1 0 1
0 0 0 1	→	0 0 0 0 1 1 0 1 1	1 0 0 1	→	1 0 1 0 1 1 1 1 0
0 0 1 0	→	0 0 0 1 0 1 1 0 1	1 0 1 0	→	1 0 1 1 0 1 0 0 0
0 0 1 1	→	0 0 0 1 1 0 1 1 0	1 0 1 1	→	1 0 1 1 1 0 0 1 1
0 1 0 0	→	0 1 1 0 0 0 0 1 1	1 1 0 0	→	1 1 0 0 0 0 1 1 0
0 1 0 1	→	0 1 1 0 1 1 0 0 0	1 1 0 1	→	1 1 0 0 1 1 1 0 1
0 1 1 0	→	0 1 1 1 0 1 1 1 0	1 1 1 0	→	1 1 0 1 0 1 0 1 1
0 1 1 1	→	0 1 1 1 1 0 1 0 1	1 1 1 1	→	1 1 0 1 1 0 0 0 0

odd parity (4x5)

0	1	0	1	1
1	0	1	1	0
0	1	1	0	1
0	1	1	1	0 ~~1~~

Even parity

0	1	0	1	0
1	0	1	1	1
0	1	1	0	0
1	0	0	0	1

Topic : Hamming Code

→ Single bit error-correcting code

→ Both transmitter and receiver must agree on same parity
[either "Even Parity" or "Odd Parity"]

Topic : Hamming Code

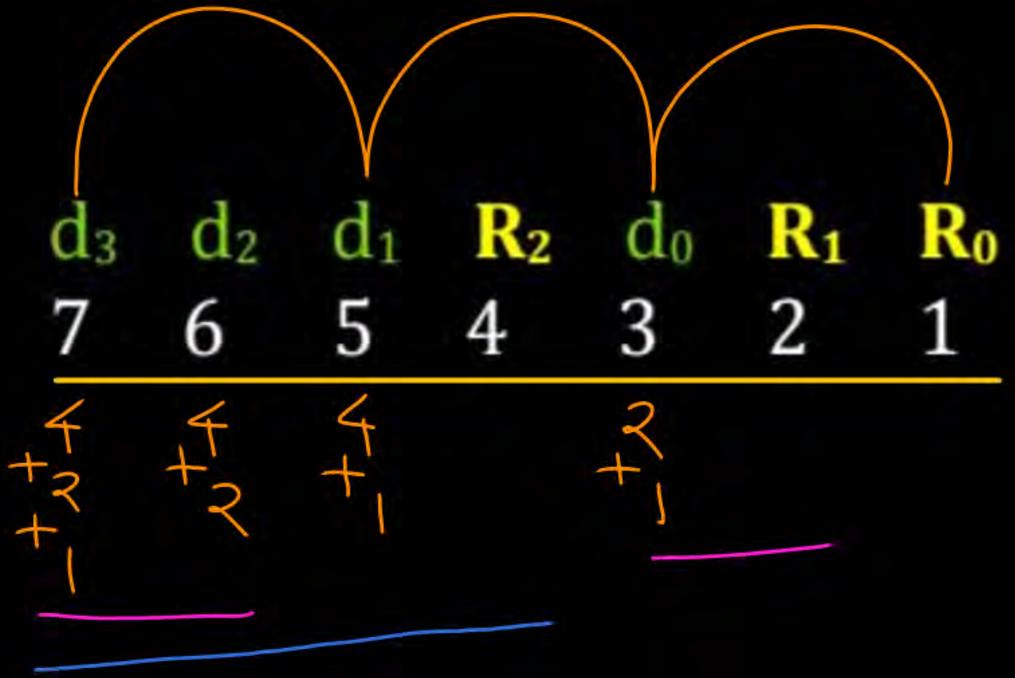
→ Parity bit (R_i) placed at position = 2^i
[where i = 0, 1, 2, 3 . . .]

Data ⇒ $d_3 d_2 d_1 d_0$

d_3	d_2	d_1	R_2	d_0	R_1	R_0
7	6	5	4	3	2	1
			2^2		2^1	2^0

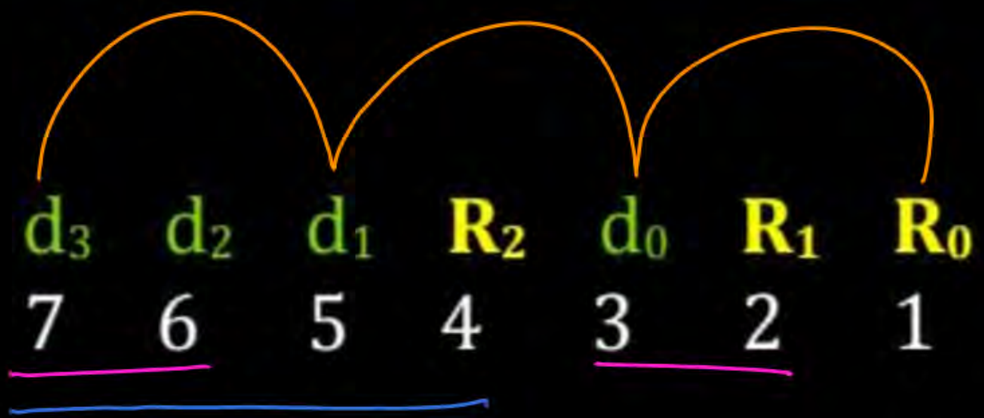
→ First parity bit placed at least significant bit position.
R_0

Topic : Hamming Code



Topic : Hamming Code

→ Even Parity



$R_0 = d_0 \oplus d_1 \oplus d_3$ [1 = 3, 5, 7]

$R_1 = d_0 \oplus d_2 \oplus d_3$ [2 = 3, 6, 7]

$R_2 = d_1 \oplus d_2 \oplus d_3$ [4 = 5, 6, 7]

Example 1 :-

#Q. Consider hamming code error control technique which uses 'even parity' and first parity bit placed at least significant bit position. The four bits data '0101' (placed from most significant bit to least significant bit position) encoded as :

Solution 1 :- (Even Parity)

Data → 0101

$R_0 = 1$

$R_1 = 0$

$R_2 = 1$

Solution 1 :- (Even Parity)

0	1	0	1	1	0	1
7	6	5	4	3	2	1

Ans : 0101101

Hamming Code
Transmitted Code

Example 2 :-

#Q. Consider hamming code error control technique which uses 'even parity' and first parity bit placed at least significant bit position. The four bits data '1010' (placed from most significant bit to least significant bit position) encoded as :

Solution 2 :- (Even Parity)

Data → 1010

$R_0 = 0$

$R_1 = 1$

$R_2 = 0$

Solution 2 :- (Even Parity)

1	0	1	0	0	1	0
7	6	5	4	3	2	1

Ans : 1010010

Topic : Hamming Code

→ Parity bit (R_i) placed at position = 2^i

[where i = 0, 1, 2, 3 . . .]

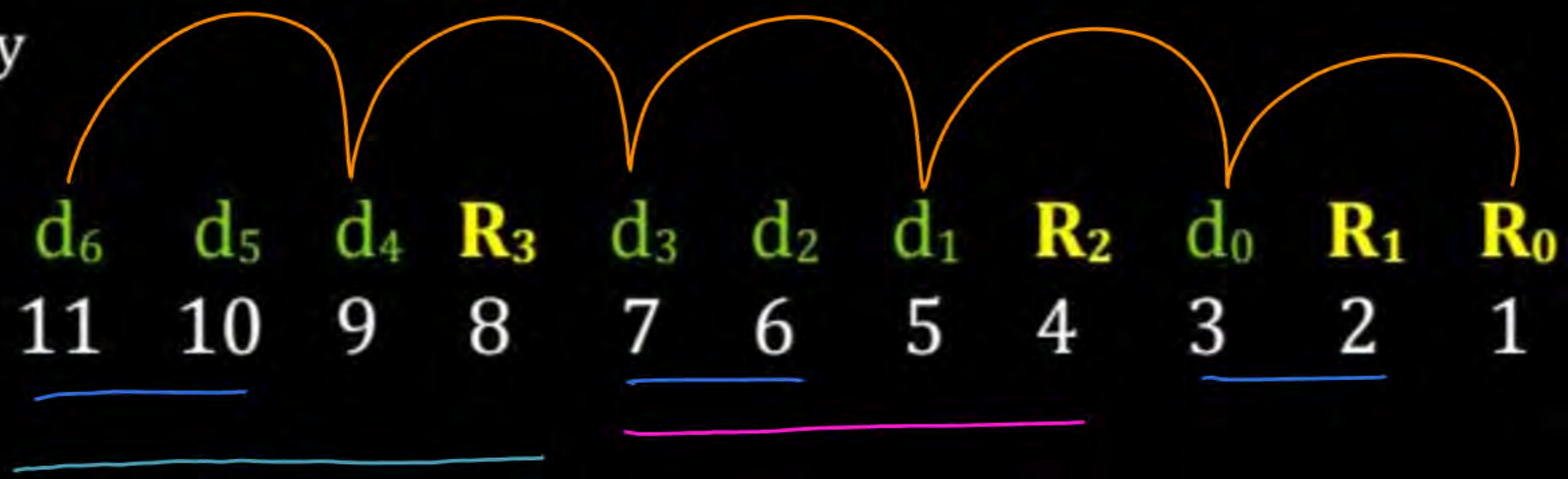
d_6	d_5	d_4	R_3	d_3	d_2	d_1	R_2	d_0	R_1	R_0
11	10	9	8	7	6	5	4	3	2	1

8 + 2 + 1 | 8 + 2 | 8 + 1 | 2^3

→ First parity bit placed at least significant bit position.

Topic : Hamming Code

→ Even Parity



$R_0 = d_0 \oplus d_1 \oplus d_3 \oplus d_4 \oplus d_6$

$R_1 = d_0 \oplus d_2 \oplus d_3 \oplus d_5 \oplus d_6$

$R_2 = d_1 \oplus d_2 \oplus d_3$

$R_3 = d_4 \oplus d_5 \oplus d_6$

[1 = 3, 5, 7, 9, 11]

[2 = 3, 6, 7, 10, 11]

[4 = 5, 6, 7]

[8 = 9, 10, 11]

Example 3 :- (H.W.)

#Q. Consider hamming code error control technique which uses 'even parity' and first parity bit placed at least significant bit position. The seven bits data '1101101' (placed from most significant bit to least significant bit position) encoded as :

2 mins Summary

Topic Two-dimension (2D) Parity

Topic Hamming Code

THANK - YOU